

Ayanesh Lal

+33-0780658734 • ayanesh88@gmail.com • <https://ayaneshlal.github.io/>

EDUCATION

Université Sorbonne Paris Nord [Coursework in Sorbonne Université & Paris Cité]	Paris
M2 Mathematics Research	2025-2026
CGPA: 14.33/20 (Mention: Bien)	
Université Sorbonne Paris Nord	Villetaneuse
M1 in Pure and Applied Mathematics	2024-2025
CGPA: 14.23/20 (Mention: Bien)	
Shiv Nadar Institution of Eminence	Greater Noida
Bachelor of Science (Research) in Mathematics	2020-2024
CGPA: 8.3/10	

FELLOWSHIPS/SCHOLARSHIPS

I'EUR PNGS M&CS Grant - Université Sorbonne Paris Nord	September 2024 – June 2026
Grant awarded based on academic performance during undergraduate degree	

RESEARCH EXPERIENCE

M2 Thesis	Paris
Topic: Linear wave equations on Schwarzschild-de Sitter spacetimes	April 2026-August 2026
Principal Advisor: Dr. Pascal Millet	
Co-advisor: Dr. Volker Schlue	
Conducted a detailed spectral and microlocal study of linear wave equations on Schwarzschild-de Sitter black-hole spacetimes, focusing on trapping, radial point estimates and Fredholm property.	
Thesis Report	
Undergraduate Research Scholar	Greater Noida
Research Topic: Representation Theory of Finite and Compact Groups	August 2023-May 2024
Principal Advisor: Dr. Amber Habib	
A year-long research project done as a part of the course MAT 499 where I studied about representations of finite and compact groups. The thesis concluded with a proof of the different Peter Weyl theorems on compact groups. I submitted a Thesis Report and gave a Poster Presentation for the same at the end of my final semester.	

SEMINARS

Université Sorbonne Paris Nord	Villetaneuse
1. Probability Laws about Jacobi Theta, Riemann Zeta functions and Brownian Excursions	2025-2026
A twenty minute talk given as a part of the seminar series 'Combinatorial Probability' where I presented various probability results related to the connections between Jacobi Theta, Riemann Zeta functions and Brownian Excursions. Slides	
2. Wasserstein Stability for Lipschitz Functions	2024-2025
A seventy five minutes talk given as a part of the seminar series 'GT Analyse topologique de données' where I presented David Cohen-Steiner's paper "Lipschitz functions have Lp-stable persistence". Handwritten Notes	
Shiv Nadar Institution of Eminence	Greater Noida
1. Bilinear Forms	2023
A thirty-minute seminar given as a part of the seminar giving course (MAT 390) where I introduced the concept of bilinear forms and talked about its various properties. The reference used was Algebra by Artin. Handwritten notes	
2. Symmetric Forms	2023
A one-hour seminar given as a part of the seminar giving course (MAT 390) where I defined what a symmetric form is, talked about a few properties and went on to prove Sylvester's law for symmetric bilinear forms. The reference used was Algebra by Artin. Slides	

PROJECTS

Université Sorbonne Paris Nord

Villetaneuse

Results in Harmonic Analysis

2024-2025

- A mandatory project done under the guidance of Dr. Emmanuel Schenck where we did a study on some specific results in Harmonic Analysis including looking at the behaviour of convolution operators where the kernel consists of an oscillatory part. [Submitted Report](#)

Université Sorbonne Paris Nord

Villetaneuse

Spectral Theorem of Self Adjoint and Unitary Operators

2024-2025

- A mandatory project done under the guidance of Dr. Hakim Boumaza where we looked at the proof of the Spectral Theorem of Self Adjoint and Unitary operators done through the Cayley Transform. [Submitted Report](#)

Indian Institute of Technology

Kanpur

Takeya's Needle Problem and Finite Field Conjecture

2023

- A summer study project done under the guidance of Dr. Parasuram Mohanty where we looked at Takeya's needle problem, self-similar shapes, Takeya's conjecture on the Hausdorff and Minkowski dimensions of Besicovitch sets and finally the statement and proof of Takeya's finite field conjecture. Properties of Hausdorff measure, self-similar shapes and space filling curves were discussed with the professor and a written report on construction of Besicovitch set and finite field Takeya conjecture was submitted. [Submitted Report](#)
References used were Real Analysis by Elias M. Stein and Rami Shakarchi, Fractal Geometry by Kenneth Falconer and Polynomial Methods in Combinatorics by Larry Guth

TALKS

MathXChange, An Indian Statistical Institute x SNIoE Talk Series

Greater Noida

1. Takeya's Needle Problem

October 2023

A hybrid talk given in the MathXChange talk series between ISI Bangalore and SNIoE. This talk was aimed at introducing Takeya's Needle Problem and providing a unique construction of the Besicovitch set using the Cantor set of dissection half. [Handwritten Notes](#)

TEACHING EXPERIENCE

1. **Tutor**

September 2023 – December 2023

Learning and Academic Support Centre, Shiv Nadar Institution of Eminence

- LASC Tutor for Calculus I (MAT 101)

- Tutored a group of 45 freshers and sophomore students that required academic support in Calculus I through a series of weekly lectures and tutorials.

AWARDS, ACHIEVEMENTS AND EXTRACURRICULAR ACTIVITIES

1. DUGC Student Representative for Mathematics
Helped the DUGC decide courses for 2 semesters along with guiding junior students through their academic work
2023-2024
2. Member of the Mathematics Society, SNIoE
2020-2024
3. Shiv Nadar Institution of Eminence Football Team
I played for the SNIoE football team
2021-2024
4. Award for Academic Excellence, Delhi Public School
Received scholar badge for obtaining over 91% grade in an academic year.
2008,2009,2011-2018,2020
5. Delhi Public School Football Team
Captained the football team for two years, won four tournaments, two player of the tournament awards and one sportsperson of the year award
2015-2020
6. Taek Won Do
Received a silver medal in the 4th Poomse District Taekwondo Championship held in Kanpur in the singles, duo and trio categories and attained black belt in 2015
2009-2015

SKILLS

1. Software:
MS Office and Latex
2. Multimedia:
Sony Vegas Pro, Adobe After Effects, Audacity and Adobe Premiere Pro
3. Languages:
English, Oriya, Hindi and French(B1)